Video Transcript - Part 11

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Hey everyone, welcome to part eleven. In this video we're looking at Amazon SageMaker, the service AWS built to handle the entire machine learning workflow in one place. Everything we've covered so far, data preparation, features, and processing, comes together here, and it's a service you'll see throughout the rest of this course.

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Here's the plan. We'll start with what SageMaker is, then follow the machine learning lifecycle from preparing data, to training, to deploying and monitoring a model. After that, we'll look at the shortcuts SageMaker offers for a faster start, and finish with the key benefits.

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So what is Amazon SageMaker? It's a fully managed service for building, training, and deploying machine learning models at scale. Fully managed means AWS runs the servers, the scaling, and the infrastructure, so you can focus on your data and your models instead of on machines. One quick note on naming: in December 2024, AWS renamed the service Amazon SageMaker AI, and the name SageMaker now describes a broader platform for data, analytics, and AI. You'll see both names used, and the machine learning features we're covering are the same.

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The easiest way to understand SageMaker is to follow the machine learning lifecycle. First you prepare your data. Then you build, which means exploring the data and writing code. Next you train a model, and tune it to improve performance. Then you deploy it so applications can get predictions. And finally you monitor it in production, because models can get worse over time. SageMaker has tools for every one of these stages.

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Let's start with preparing and building. SageMaker Studio is the web based environment where you write code, run notebooks, and manage everything in one place. Data Wrangler lets you import, explore, and transform data visually, which covers things like handling missing values, encoding, and scaling that we just discussed. Feature Store is a central place to save and reuse features, so different teams don't rebuild the same ones. And Ground Truth helps you label data, using human reviewers and automation, when you need labeled examples for supervised learning.

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Next, training and tuning. In SageMaker, you start a training job and tell it where your data lives in Amazon S3. It spins up the compute, trains the model, saves the result back to S3, and shuts the machines down. You can use built in algorithms like XGBoost, bring your own code with frameworks like TensorFlow, PyTorch, or scikit-learn, or use your own container. Automatic model tuning runs many training jobs with different hyperparameter settings to find the best combination for you. And managed spot training can cut your training costs significantly by using spare compute capacity.

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Once the model is trained, you deploy it, and SageMaker gives you four main options. Real time endpoints keep a model running for low latency predictions, ideal for things like fraud checks during a purchase. Serverless inference scales up and down automatically, even to zero, which suits traffic that comes in bursts. Asynchronous inference queues requests, so it handles large payloads and long processing times. And batch transform runs predictions over a whole dataset at once, with no endpoint to keep running. Picking the right one comes down to latency needs, traffic patterns, and cost.

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After deployment, the work isn't finished. Model Monitor watches live predictions and alerts you to problems like data drift, where incoming data starts to look different from your training data. Clarify helps detect bias and explain why a model made a prediction. Model Registry keeps track of model versions and approvals, so you control what goes to production. And SageMaker Pipelines lets you automate the whole workflow, from data preparation to deployment, as a repeatable process. That kind of automation is what people call MLOps.

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Not everyone wants to write everything from scratch, so SageMaker offers shortcuts. Autopilot is AutoML. Give it a dataset and a target, and it tries different algorithms and settings, and shows you what it did. JumpStart is a hub of pretrained models, including foundation models, and ready made solutions you can deploy in a few clicks. And Canvas is a no code interface, so business analysts can build predictions through a visual point and click experience.

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So why use SageMaker? It's fully managed, so there are no servers to maintain. It scales from a small experiment to large distributed training. You pay only for what you use, and training machines shut down when jobs finish. It integrates with the rest of AWS, including S3 for data, IAM for permissions, and CloudWatch for logs and metrics. And it covers the whole lifecycle, so you don't need to stitch together separate tools.

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So to recap, SageMaker is AWS's fully managed platform for the whole machine learning lifecycle. You can prepare data, train and tune models, deploy them in several ways, and monitor them in production, with shortcuts like Autopilot, JumpStart, and Canvas along the way. In the coming videos, we'll go deeper into how these pieces work together. Thanks for watching, and I'll see you in the next one.